Software Test Plan (STP): Event Management System

Project: Event Management System (EMS)

Version: 2.0

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Status: Final, with test execution results (end of Sprint 2)

Course: UE24CS341A Software Engineering, PES University. Group 7 (to confirm)

Revision history

Version	Date	Author	Change summary
1.0	12-10-2026	Team	Initial test plan with 53 test cases and traceability to the SRS
1.1	13-10-2026	Varun Rao	Test schedule changed from calendar dates to sprint phases
2.0	10-11-2026	Team	Test cases updated to reflect the sprint changes and executed on the final Sprint 2 build; Actual Result and Result columns completed; system validation results added in section 14.1

1. Introduction

Purpose: This document defines the test plan for the Event Management System (EMS) v1.0. It describes the objectives, scope, strategy, environment, schedule and responsibilities for testing, and links every test to a requirement in the SRS.

Scope: Testing covers the four features of EMS: authentication and role management, event creation and management, event registration and ticketing, and check-in, notifications and reports. It also covers the non-functional requirements (performance, reliability, usability, concurrency, maintainability, compatibility) and the security requirements of the SRS.

References: EMS SRS v1.0 (sections 4, 5.1, 6 and 8), EMS Software Architecture and Design Specification (in preparation), project guidelines and the IEEE test plan template.

Definitions:

Term	Meaning
EMS	Event Management System
SRS / SAD / STP / RTM	Software Requirements Specification / Software Architecture and Design Specification / Software Test Plan / Requirements Traceability Matrix
TC	Test case
FR / NFR / SR	Functional / Non-functional / Security requirement
RBAC	Role-Based Access Control
JWT	JSON Web Token
UAT	User acceptance testing

2. Test Items

- F1 Authentication and role management module (registration, login, lockout, logout, RBAC, user administration, profile)
- F2 Event management module (create, edit, publish, cancel, categories, browse, search and filter)
- F3 Registration and ticketing module (register, waitlist, cancel, promotion, tickets with QR code, registration lists)
- F4 Check-in, notifications and reports module (check-in, emails, reminders, in-app notifications, reports, CSV export, admin dashboard)
- React frontend, Node.js/Express REST API and PostgreSQL database as an integrated system

3. Features to be Tested

Requirement ID	Feature	Test cases
EVT-F-001	Register an account	TC-AUTH-01, 02, 10
EVT-F-002	Login with a 60-minute session token	TC-AUTH-03, 09
EVT-F-003	Account lockout after 5 failed logins	TC-AUTH-04
EVT-F-004	Logout invalidates the session	TC-AUTH-05
EVT-F-005	Role-based access control (HTTP 403)	TC-AUTH-06, TC-EVT-10
EVT-F-006	Admin user management	TC-AUTH-07
EVT-F-007	Profile and password change	TC-AUTH-08
EVT-F-010	Create event	TC-EVT-01, 10
EVT-F-011	Event data validation	TC-EVT-02, 03

Requirement ID	Feature	Test cases
EVT-F-012	Edit own event, capacity rule	TC-EVT-04
EVT-F-013	Publish event, Draft hidden	TC-EVT-05
EVT-F-014	Cancel event	TC-EVT-06
EVT-F-015	Category management	TC-EVT-07
EVT-F-016	Browse, search, filter, paginate	TC-EVT-08, 09
EVT-F-020	Register for an event	TC-REG-01, 02
EVT-F-021	No duplicate registration	TC-REG-03
EVT-F-022	Waitlist	TC-REG-04
EVT-F-023	Cancel own registration	TC-REG-05, 10
EVT-F-024	Automatic waitlist promotion	TC-REG-06
EVT-F-025	My registrations list	TC-REG-07
EVT-F-026	Ticket with QR code	TC-REG-08
EVT-F-027	Organizer registration list	TC-REG-09
EVT-F-030	Check-in	TC-CHK-01
EVT-F-031	Check-in rejection cases	TC-CHK-02, 03
EVT-F-032	Notification emails	TC-CHK-04
EVT-F-033	24-hour reminder email	TC-CHK-05
EVT-F-034	In-app notifications	TC-CHK-06
EVT-F-035	Attendance report	TC-CHK-07, 10
EVT-F-036	CSV export	TC-CHK-08
EVT-F-037	Admin dashboard	TC-CHK-09
EVT-NF-001	95% of requests within 2 s at 100 users	TC-NFR-01
EVT-NF-002	99% monthly availability	TC-NFR-02
EVT-NF-003	Usability: 3 clicks, 360 px to 1920 px	TC-NFR-03
EVT-NF-004	No overbooking under 500 simultaneous attempts	TC-NFR-04
EVT-NF-005	70% unit test coverage and quality gate	TC-NFR-05
EVT-NF-006	Browser compatibility	TC-NFR-06
EVT-SR-001 to EVT-SR-007	Security requirements	TC-SEC-01 to TC-SEC-07 (see section 5.1)

4. Features Not to be Tested

- Online payments and paid ticketing (out of scope in the SRS)
- Venue booking and seat maps (out of scope)
- Native mobile applications (out of scope)
- Integration with third-party calendars or social networks (out of scope)
- Internal behaviour of the external SMTP email service (only that EMS hands over the correct email is tested; a mock mail server is used)
- Third-party libraries and framework internals (React, Express, PostgreSQL), which are assumed to be tested by their maintainers

5. Test Approach / Strategy

5.0 Test levels

Level	What is tested	How
Unit	Validation rules, calculations (attendance rate, waitlist order), password and token helpers	Jest for the Node.js backend and React Testing Library for the frontend, run by GitHub Actions

Level	What is tested	How
		on every pull request. Target: at least 70% line coverage (EVT-NF-005)
Integration	REST API with the PostgreSQL test database, role checks, email hand-over to a mock SMTP server	Jest with Supertest and Postman collections
System	End-to-end flows through the web interface on the supported browsers	Manual execution of the test cases in tc-01 to tc-04, with screenshots
Acceptance	All acceptance criteria (AC-EVT-...) in the SRS and the exit criteria in SRS 6.1	Manual walkthrough by the team at the end of Sprint 2

5.0.1 Test types

- **Functional testing:** every functional requirement has at least one test case, with both valid and invalid input.
- **Regression testing:** the automated unit and integration suites run in the CI pipeline on every pull request, and the manual smoke cases (TC-AUTH-03, TC-EVT-01, TC-REG-01, TC-CHK-01) are repeated before each sprint review.
- **Performance testing:** JMeter with 100 concurrent users (TC-NFR-01).
- **Concurrency testing:** a script sending 500 simultaneous registrations (TC-NFR-04).
- **Usability and responsive testing:** walkthrough with 5 users and layout checks at 360 px and 1920 px (TC-NFR-03).
- **Compatibility testing:** latest two versions of Chrome, Firefox and Edge (TC-NFR-06).
- **Security testing:** see section 5.1.

5.0.2 Test design techniques

- **Boundary value analysis:** event capacity 0, 1, 10000 and 10001; password length 7 and 8; lockout at the 5th and 6th failed login; rate limit at the 100th and 101st request.
- **Equivalence partitioning:** valid and invalid classes for email, dates, capacity, ticket status and user roles.
- **State-based testing:** registration status changes (Confirmed, Waitlisted, Cancelled, Checked-in) and event status changes (Draft, Published, Cancelled).
- **Error guessing:** double clicks, repeated submissions and expired sessions.

5.0.3 Defect severity

Severity	Meaning	Example
Critical	System unusable or data or security breach	Overbooking, login bypass
High	A main feature fails with no workaround	Registration cannot be confirmed
Medium	A feature fails but a workaround exists	Filter shows wrong page count
Low	Cosmetic or minor	Misaligned label

5.0.4 Entry and exit criteria

Entry criteria: a stable build is deployed on the test environment, test data is loaded, the CI pipeline is green and the test cases for the feature are reviewed.

Exit criteria (from SRS 6.1): all High-priority requirements are implemented and verified; no critical or high-severity defect is open; all non-functional and security requirements meet their targets; every requirement is mapped to at least one passed test case in the RTM.

6. Test Environment

Hardware: team laptops (minimum 8 GB RAM) running the application locally, with phones or a browser window resized to 360 px for responsive checks.

Software:

Item	Version or detail
Operating system	Windows 10 or 11 (development); Linux with Docker for the CI pipeline
Node.js	LTS version used by the project
Database	PostgreSQL test database, reset with seed data before each run
Browsers	Latest two versions of Google Chrome, Mozilla Firefox and Microsoft Edge
Mock mail server	A local SMTP test server (for example MailHog) to read the emails EMS sends

Tools:

Purpose	Tool
Unit and integration tests	Jest, Supertest, React Testing Library
API testing	Postman
Performance and concurrency	JMeter (or k6)
Security scanning	OWASP ZAP, a TLS scanner (SSL Labs or testssl.sh), npm audit
Static analysis and coverage	SonarQube
CI/CD	GitHub Actions
Defect and task tracking	GitHub Issues and GitHub Projects

Test data: one Admin, two Organizers and at least six Attendee accounts; a set of Draft, Published and Cancelled events across several categories and dates; one event with capacity 2 for waitlist tests and one with capacity 100 for the concurrency test; 500 generated attendee accounts for TC-NFR-04. All data is dummy data.

5.1 Security Validation

Security validation checks that the security requirements EVT-SR-001 to EVT-SR-007 and the objectives SO-1 to SO-3 in the SRS are met.

Security check	Method and tool	Requirement	Test case
Passwords are stored as salted bcrypt hashes and never logged	Database inspection and log review	EVT-SR-001	TC-SEC-01
Tokens expire after 60 minutes and tampered, expired or missing tokens are rejected	Manual API calls with Postman	EVT-SR-002	TC-SEC-02
Injection and script attacks in all input fields	Fuzzing with SQL injection and script strings and an OWASP ZAP scan	EVT-SR-003	TC-SEC-03
HTTPS with TLS 1.2 or higher, HTTP redirected	TLS scan with SSL Labs or testssl.sh	EVT-SR-004	TC-SEC-04
Users cannot read other users' registrations or tickets	Changing IDs in requests (access control test)	EVT-SR-005	TC-SEC-05

Security check	Method and tool	Requirement	Test case
Security events are written to the audit log	Perform each action, inspect the log	EVT-SR-006	TC-SEC-06
Request and login rate limiting returns HTTP 429	Script sending 101 requests and 11 logins per minute	EVT-SR-007	TC-SEC-07
Role-based access control (HTTP 403 for forbidden actions)	Negative tests for every role	EVT-F-005	TC-AUTH-06, TC-EVT-10
Vulnerable dependencies and secrets in the repository	npm audit, SonarQube analysis and a repository check for committed secrets	EVT-NF-005	TC-NFR-05

Penetration-style checks: the authentication flow (login, lockout, token handling) is attacked by repeated login attempts, token modification and role escalation tries (an Attendee calling Organizer and Admin actions).

Reporting: every security defect is raised as a GitHub issue labelled `security` with severity, and a critical or high security defect blocks the release (see exit criteria).

7. Test Schedule

The schedule is expressed in project phases and sprints instead of calendar dates, so it stays valid if the faculty changes the dates.

Milestone	Phase
Test plan and test case design	Before Sprint 1 (this document)
Test environment and CI setup	Start of Sprint 1
Sprint 1 test execution (first half of each feature)	Sprint 1
Sprint 2 test execution and regression	Sprint 2
Final results, Actual and Result columns completed, test summary	End of Sprint 2, before the code freeze

8. Test Deliverables

- Test Plan (this document)
- Test cases with Actual Result and Result filled in (tc-01 to tc-04)
- Automated unit and integration test code and the coverage report
- Test data and seed scripts
- Test execution evidence (screenshots and logs)
- Defect reports (GitHub Issues)
- Security scan reports (OWASP ZAP and TLS scan)
- Test summary report

9. Roles and Responsibilities

Role	Name	Responsibility
Test Lead	Varun Rao	Prepares and maintains the plan, assembles the traceability matrix and the test summary report
Test Engineer, F1	Varun Rao	Designs and runs TC-AUTH, TC-NFR-01, TC-SEC-01, TC-SEC-02
Test Engineer, F2	Thanishq Raam Patil	Designs and runs TC-EVT, TC-NFR-02, TC-NFR-06, TC-SEC-03

Role	Name	Responsibility
Test Engineer, F3	Vishruta	Designs and runs TC-REG, TC-NFR-03, TC-NFR-04, TC-SEC-04, TC-SEC-05
Test Engineer, F4	Nithin Kumar M	Designs and runs TC-CHK, TC-NFR-05, TC-SEC-06, TC-SEC-07
Repository owner and reviewer	Vishruta	Reviews and merges pull requests, owns the CI pipeline
Developer	Each member	Fixes defects found in their own feature
Approver	Course faculty	Approves the test results

10. Risks and Mitigation

Risk	Mitigation
Short sprints leave little time to test	Write test cases before coding, automate unit tests early and run them in CI
A feature depends on another member's feature (for example check-in needs tickets)	Agree the data and API shapes early and use seeded test data
Test environment or database breaks	Keep seed and reset scripts in the repository so the environment can be rebuilt quickly
Email service is unavailable or slow	Use a mock SMTP server for tests
Concurrency and performance tests need many accounts and a stable machine	Generate test accounts by script and run the load test on a quiet machine
Team member is unavailable	Every test case is reviewed by a second member, who can run it

11. Assumptions and Dependencies

- A stable build is available at the start of each sprint
- Test data and the mock SMTP server are available before execution
- The SRS requirements and acceptance criteria are the baseline for the expected results
- All event times use one configured time zone
- GitHub Actions is enabled for the repository

12. Suspension and Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours
- The build is so unstable that more than 30% of the planned test cases are blocked
- A critical defect blocks the login or registration flow

Resume testing when:

- The blocking defects are fixed and a new build is deployed
- The environment is stable again and a smoke test (TC-AUTH-03, TC-EVT-01, TC-REG-01, TC-CHK-01) passes

13. Test Case Management and Traceability

Test cases are listed with their results in Appendix A and stored in `docs/test-plan/` and use the ID prefixes defined in SRS 6.2. Every requirement maps to at least one test case, and the table below is the source for section 8 (RTM) of the SRS.

Requirement ID	Test cases
EVT-F-001	TC-AUTH-01, TC-AUTH-02, TC-AUTH-10
EVT-F-002	TC-AUTH-03, TC-AUTH-09
EVT-F-003	TC-AUTH-04
EVT-F-004	TC-AUTH-05
EVT-F-005	TC-AUTH-06, TC-EVT-10
EVT-F-006	TC-AUTH-07
EVT-F-007	TC-AUTH-08
EVT-F-010	TC-EVT-01, TC-EVT-10
EVT-F-011	TC-EVT-02, TC-EVT-03
EVT-F-012	TC-EVT-04
EVT-F-013	TC-EVT-05
EVT-F-014	TC-EVT-06
EVT-F-015	TC-EVT-07
EVT-F-016	TC-EVT-08, TC-EVT-09
EVT-F-020	TC-REG-01, TC-REG-02
EVT-F-021	TC-REG-03
EVT-F-022	TC-REG-04
EVT-F-023	TC-REG-05, TC-REG-10
EVT-F-024	TC-REG-06
EVT-F-025	TC-REG-07
EVT-F-026	TC-REG-08
EVT-F-027	TC-REG-09
EVT-F-030	TC-CHK-01
EVT-F-031	TC-CHK-02, TC-CHK-03
EVT-F-032	TC-CHK-04
EVT-F-033	TC-CHK-05
EVT-F-034	TC-CHK-06
EVT-F-035	TC-CHK-07, TC-CHK-10
EVT-F-036	TC-CHK-08
EVT-F-037	TC-CHK-09
EVT-NF-001	TC-NFR-01
EVT-NF-002	TC-NFR-02
EVT-NF-003	TC-NFR-03
EVT-NF-004	TC-NFR-04
EVT-NF-005	TC-NFR-05
EVT-NF-006	TC-NFR-06
EVT-SR-001	TC-SEC-01
EVT-SR-002	TC-SEC-02
EVT-SR-003	TC-SEC-03
EVT-SR-004	TC-SEC-04
EVT-SR-005	TC-SEC-05
EVT-SR-006	TC-SEC-06
EVT-SR-007	TC-SEC-07

Test case count per feature: F1 has 10 (TC-AUTH), F2 has 10 (TC-EVT), F3 has 10 (TC-REG) and F4 has 10 (TC-CHK), plus 6 non-functional and 7 security test cases, which is 53 in total.

14. Test Metrics and Reporting

Metrics collected:

- Percentage of test cases executed
- Percentage passed and failed
- Defect count by severity and by feature (defect density)
- Defect aging (days from raised to closed)
- Requirement coverage (requirements with at least one passed test case)
- Unit test line coverage (target 70% or higher)

Reports:

- Status of executed test cases at the end of each sprint
- Final test summary report with the Actual Result and Result columns completed

14.1 Test Execution Results

Build tested: `main` at the end of Sprint 2 (commit 2fe85c5), started with Docker Compose on a fresh database.

Date: 10 Nov 2026.

Scope: all 53 test cases of the Test Plan (40 functional, 6 non-functional, 7 security), run at system and integration level against the complete system.

Detailed results: Appendix A of this document.

Related reports: unit test report, code and branch coverage report, and one report per non-functional and security requirement, in docs/test-reports.

Results by feature

Feature	Owner	Test cases	Pass	Pass with note	Partially executed
F1 Authentication and role management	Varun Rao	TC-AUTH-01 to 10, TC-NFR-01, TC-SEC-01, TC-SEC-02	13	0	0
F2 Event creation and management	Thanishq Raam Patil	TC-EVT-01 to 10, TC-NFR-02, TC-NFR-06, TC-SEC-03	11	1	1
F3 Registration and ticketing	Vishruta	TC-REG-01 to 10, TC-NFR-03, TC-NFR-04, TC-SEC-04, TC-SEC-05	14	0	0
F4 Check-in, notifications and reports	Nithin Kumar M	TC-CHK-01 to 10, TC-NFR-05, TC-SEC-06, TC-SEC-07	12	1	0
Total		**53**	**50**	**2**	**1**

No test case failed. Requirement coverage: all 43 requirements in the SRS have at least one executed test case.

Results with a note

Test case	Result	Note
TC-NFR-02	Pass (short test period)	100% uptime over the test period, and Docker restarted the backend by itself after an engine restart. A month-long measurement needs a deployed server.
TC-NFR-05	Pass with deviation	Coverage 86.79% lines and 78.64% branches (target 70% lines) and the CI gate works. SonarQube analysis was not run; the quality gate is enforced in CI by ESLint (0 errors) and the coverage threshold.
TC-NFR-06	Partially executed (Chrome only)	All main flows pass in Chrome. Firefox and Edge were not tested.

Test cases updated to reflect the sprints

Test case	Change	Reason
TC-EVT-01, TC-EVT-07, TC-EVT-09	Test data uses the seeded category names (Technology, Music, "Workshop Series")	"Technical" does not exist and "Workshop" is already seeded, so adding it again is correctly refused as a duplicate
TC-EVT-09	Expected result: cancelled and past events are not listed; a cancelled event's own page shows Cancelled	Search lists only upcoming published events (Sprint 1 review, deviation 7)
TC-NFR-01	Steps: team load-test script (`backend/scripts/load-test.js`) instead of a JMeter plan	Runs inside the project's Docker setup with no extra tool; measures the same 95th percentile
TC-SEC-04	Steps: `openssl s_client` with each TLS version instead of an SSL Labs or testssl.sh scan	SSL Labs can only scan public servers; the system runs locally
TC-NFR-05	Steps and expected result: coverage gate and ESLint in CI instead of a SonarQube quality gate	SonarQube analysis was dropped; the CI checks enforce the same rule on every pull request

Test levels

Level	How it was tested	Count
Unit	Jest (backend) and Vitest (frontend), run in GitHub Actions on every pull request	205 backend, 36 frontend (unit test report)
Integration	API calls against the running backend and PostgreSQL database	test cases marked (I)
System	Full system through the web interface over HTTPS in Chrome	test cases marked (S)

Non-functional and security measurements

Requirement	Result	Report
EVT-NF-001 Performance	95th percentile 138 ms with 100 users for 5 minutes	nf-001-performance.md
EVT-NF-002 Availability	100% over the test period, automatic restart	nf-002-availability.md
EVT-NF-003 Usability	3 clicks for a visitor, no sideways scrolling from 360 px to 1920 px	nf-003-usability.md
EVT-NF-004 No overbooking	3 runs of 500 requests for 100 seats: exactly 100 confirmed each time	nf-004-concurrency.md
EVT-NF-005 Coverage	86.79% lines, 78.64% branches	coverage-report.md

Requirement	Result	Report
EVT-NF-006 Browsers	Chrome passes; Firefox and Edge not tested	nf-006-browser-compatibility.md
EVT-SR-003 Input validation	Injection and script strings handled safely	sr-003-input-validation.md
EVT-SR-004 HTTPS	TLS 1.0/1.1 refused, 1.2/1.3 accepted, HTTP redirected	sr-004-https.md

Conclusion

The system meets its functional, security and measurable non-functional requirements on the final Sprint 2 build. The open points are a month-long availability measurement, which needs a deployed server, and testing in Firefox and Edge.

Detailed reports for the non-functional and security requirements (how each measurement is repeated, and its results) are in the repository under docs/test-reports.

15. Approvals

Role	Name	Signature / Date
Test Lead	Varun Rao	
Repository owner	Vishruta	
Course faculty		

Appendix A. Test Cases and Execution Results

Columns follow the course format. Level in brackets in the Module column: Unit (U), Integration (I) or System (S). All test cases were executed on 10-11-2026 against the final Sprint 2 build on a fresh database with the seeded test accounts and the mock SMTP service; steps were carried out through the web interface in Chrome and through the REST API, with database queries to confirm the stored state.

A.1 F1 Authentication and Role Management

Owner: Varun Rao (PES1UG24CS520)

Covers: EVT-F-001 to EVT-F-007, EVT-NF-001, EVT-SR-001, EVT-SR-002

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-AUTH-01	F1 Authentication (S)	Register with valid details (EVT-F-001)	Visitor is on the sign-up page; email not yet used	1) Open sign-up; 2) Enter name, email, password and role; 3) Submit	Asha Rao, asha@test.com, Passw0rd1, role Attendee	Account is created, a success message is shown and the user can log in	Account created (201, role ATTENDEE); the sign-up page shows "Account created. You can log in now." and login with the new account works	Pass
TC-AUTH-02	F1 Authentication (I)	Reject a duplicate email (EVT-F-001)	An account with asha@test.com exists	1) Open sign-up; 2) Enter the same email with any other valid data; 3) Submit	asha@test.com, Passw0rd1	Message "Email already registered" is shown and no second account is created	409 EMAIL_EXISTS: "Email already registered"; still 1 account with this email	Pass
TC-AUTH-03	F1 Authentication (S)	Log in with correct credentials (EVT-F-002)	Active account exists	1) Open login; 2) Enter email and password; 3) Submit	asha@test.com, Passw0rd1	A token valid for 60 minutes is issued and the role dashboard opens	200, token issued with expiry 60 min (exp - iat = 3600 s); the browser opens the attendee dashboard (My registrations)	Pass
TC-AUTH-04	F1 Authentication (I)	Lock account after 5 failed attempts (EVT-F-003)	Active account exists and is not locked	1) Submit a wrong password 5 times; 2) Submit the correct password (6th attempt); 3) Move the lock time back by 15 minutes (or wait) and log in again	asha@test.com, wrong password Wrong123	6th attempt is refused even with the correct password; login works after 15 minutes; the failures appear in the audit log	Attempts 1-5: 401, 401, 401, 401, 401; 6th with correct password: 423 ACCOUNT_LOCKED: "Account locked after 5 failed logins. Try again in 15 minute(s)."; after moving the lock time back 15 min: 200 (login works); 6 LOGIN_FAILED entries in audit_logs	Pass
TC-AUTH-05	F1 Authentication (I)	Log out invalidates the session (EVT-F-004)	User is logged in and holds a token	1) Click log out; 2) Call a protected action with the old token	Token from TC-AUTH-03	Old token is rejected with HTTP 401	Logout 204; old token afterwards: 401 UNAUTHORIZED: "Token has been revoked"	Pass
TC-AUTH-06	F1 Authentication (I)	Role-based access is enforced (EVT-F-005)	Logged in as Attendee	1) Attempt an Organizer-only action (create event); 2) Attempt an Admin-only action (open the user list)	Attendee token	Both actions return HTTP 403 and no data changes	Create event: 403 FORBIDDEN: "You do not have permission for this action"; user list: 403 FORBIDDEN: "You do not have permission for this action"; no event saved	Pass
TC-AUTH-07	F1 Authentication (S)	Admin manages users (EVT-F-006)	Logged in as Admin; an Attendee account exists	1) Open the user list; 2) Change the Attendee role to Organizer; 3) Deactivate the account and try to log in as that user; 4) Try to deactivate the admin's own	Admin account, user asha@test.com	Role change applies at the user's next login; deactivated user cannot log in; self-deactivation is refused	User list shows 8 users; role change 200, next login returns role ORGANIZER; deactivated user login: 403 ACCOUNT_DEACTIVATED: "This account has been deactivated"; admin deactivating own account: 409	Pass

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
				account			INVALID_STATE: "You cannot deactivate your own account"	
TC-AUTH-08	F1 Authentication (S)	Update profile and change password (EVT-F-007)	User is logged in	1) Update name and phone; 2) Change password with the correct current password; 3) Change password with a wrong current password	Phone 9876543210, current Passw0rd1, new Newpass22	Profile saved; correct current password allows the change; wrong one is rejected with an error	Profile saved (name "Asha R", phone 9876543210); change with correct current password: 204, new password logs in; wrong current password: 403 FORBIDDEN: "Current password is wrong"	Pass
TC-AUTH-09	F1 Authentication (I)	Reject wrong credentials without revealing which field is wrong (EVT-F-002)	Active account exists	1) Login with a correct email and wrong password; 2) Login with an unknown email and any password	asha@test.com / Wrong123; nobody@test.com / Passw0rd1	Both attempts show the same generic error and no token is issued	Both attempts: 401 INVALID_CREDENTIALS: "Invalid email or password"; identical message, no token	Pass
TC-AUTH-10	F1 Authentication (U)	Reject invalid registration input (EVT-F-001)	Sign-up validation is available	1) Submit with password shorter than 8 characters; 2) Submit a password with no digit; 3) Submit an invalid email format; 4) Submit with an empty name	abc12; Password; asha@; empty name	Each case is rejected with a specific message and no account is created	short password: 400 "Password must have at least 8 characters"; no digit: 400 "Password must contain a letter and a digit"; bad email: 400 "Enter a valid email address"; empty name: 400 "Name is required"; no account created	Pass

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-NFR-01	F1 Performance (S)	Response time under load (EVT-NF-001)	Reference test environment with seeded data	1) Run backend/scripts/load-test.js with 100 concurrent users for 5 minutes on event list, search, event page, categories and My registrations; 2) Read the 95th percentile response time (Updated after Sprint 2.)	100 virtual users, 5 min	95th percentile is at or below 2 seconds	322,651 requests, 0 errors, median 96 ms, 95th percentile 138 ms, max 385 ms; 100% within 2 s. Login is not in the mix (bcrypt is slow by design and the login rate limit applies); registration under load is covered by TC-NFR-04	Pass

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-SEC-01	F1 Security (I)	Passwords are stored as bcrypt hashes (EVT-SR-001)	Account created in TC-AUTH-01	1) Inspect the password column in the database; 2) Search application logs and API responses for the password	asha@test.com, Passw0rd1	Only a salted bcrypt hash with cost 10 or higher is stored; the plain password appears nowhere	Stored value is a bcrypt hash "$2a$10$..." (cost 10, salted); the plain passwords appear nowhere in the server log, audit log or API responses	Pass

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-SEC-02	F1 Security (I)	Invalid session tokens are rejected (EVT-SR-002)	Valid token available	1) Call a protected action with a token with one character changed; 2) Call with an expired token; 3) Call with no token	Tampered, expired and missing token	Every call returns HTTP 401	Tampered: 401 UNAUTHORIZED: "Invalid or expired token"; expired: 401 UNAUTHORIZED: "Invalid or expired token"; missing: 401 UNAUTHORIZED: "Authentication required"	Pass

A.2 F2 Event Creation and Management

Owner: Thanishq Raam Patil (PES1UG24CS502)

Covers: EVT-F-010 to EVT-F-016, EVT-NF-002, EVT-NF-006, EVT-SR-003

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-EVT-01	F2 Event Management (S)	Create an event with valid data (EVT-F-010)	Logged in as Organizer	1) Open the event form; 2) Fill every mandatory field; 3) Save	Title Tech Fest, category Technology, venue Hall A, start in 7 days, end 2 hours later, capacity 50	Event is saved with status Draft and appears in the organizer's event list	201, saved with status DRAFT and listed under My events (category Technology used for "Technical")	Pass
TC-EVT-02	F2 Event Management (U)	Reject invalid title and dates (EVT-F-011)	Logged in as Organizer	1) Save with an empty title; 2) Save with a start date in the past; 3) Save with an end date before the start	Empty title; start yesterday; end earlier than start	Each case shows its own error and nothing is saved	empty title: 400 "Title is required"; start yesterday: 400 "Start must be in the future"; end before start: 400 "End must be after the start"; nothing saved	Pass
TC-EVT-03	F2 Event Management (U)	Reject invalid capacity (EVT-F-011)	Logged in as Organizer	1) Save with capacity 0; 2) Save with capacity 10001; 3) Save with capacity 5.5; 4) Save with capacity abc; 5) Save with capacity 1 and capacity 10000	0, 10001, 5.5, abc, then 1 and 10000	The first four are rejected with an error; the boundary values 1 and 10000 are accepted	0: 400, 10001: 400, 5.5: 400, abc: 400, 1: 201, 10000: 201 ("Capacity must be a whole number from 1 to 10000" for the rejected ones)	Pass
TC-EVT-04	F2 Event Management (I)	Edit own event and enforce capacity rule (EVT-F-012)	Own event with capacity 50 and 20 confirmed registrations; a second organizer exists	1) Change the venue and save; 2) Reduce capacity to 10; 3) Reduce capacity to 25; 4) As the second organizer edit the first organizer's event	Capacity 10 then 25	Venue change saved; capacity 10 rejected; capacity 25 saved; the other organizer gets HTTP 403	Venue change saved (200); capacity 10: 409 CAPACITY_BELOW_CONFIRMED: "Capacity cannot be less than the 20 confirmed registrations"; capacity 25: 200 saved; second organizer: 403 FORBIDDEN: "You can only edit your own events"	Pass
TC-EVT-05	F2 Event Management (S)	Publish makes an event public, Draft stays	One Draft event exists	1) Search for the event as Visitor and as Attendee; 2) Publish it as Organizer; 3)	Event Tech Fest	Draft is not visible to Visitor or Attendee; after publishing it	Before publishing: public search finds 0, attendee opening it gets 404; after publishing: search finds 1	Pass

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
		hidden (EVT-F-013)		Search again		appears in public search		
TC-EVT-06	F2 Event Management (I)	Cancel an event cancels its registrations (EVT-F-014)	Published own event with 3 Confirmed registrations	1) Cancel the event as Organizer; 2) Open the event and each registration	Event with 3 registrations	Event shows Cancelled, all 3 registrations become Cancelled and cancellation notifications are triggered	Event status CANCELLED; registrations: CANCELLED, CANCELLED, CANCELLED; 3 in-app notifications and 3 emails "Event cancelled: Cancel test"	Pass
TC-EVT-07	F2 Event Management (S)	Admin manages categories (EVT-F-015)	Logged in as Admin; a category used by an event exists (Technology)	1) Add category Workshop Series (Workshop is already a seeded category); 2) Rename it to Workshop Sessions; 3) Check it on the event form; 4) Delete a category that is in use; 5) Delete an unused category	Workshop Series, Workshop Sessions, category Technology in use	New category is selectable; deleting a category in use is refused with a message; unused one is deleted	Adding "Workshop" again is refused as a duplicate (409 "A category with this name already exists"); added "Workshop Series" (201), renamed it to "Workshop Sessions" (200) and it appears in the event form's category list; deleting Technology (in use): 409 "This category is used by 5 event(s) and cannot be deleted"; deleting the unused one: 204	Pass
TC-EVT-08	F2 Event Management (S)	Search published events by keyword (EVT-F-016)	Several published events and one Draft exist	1) Search for a keyword in a title; 2) Search for a keyword in a description; 3) Search for a keyword that matches nothing	Keywords tech, coding, zzzz	Only matching published events are shown; no match shows an empty-result message; the Draft never appears	"tech": 1 result(s) (Tech Fest); "coding" (in the description): 1 (Tech Fest); "zzzz": 0, the page shows "No events match your search"; the Draft never appears	Pass
TC-EVT-09	F2 Event Management (S)	Filter and paginate results (EVT-F-016)	More than 10 published events across categories and dates	1) Filter by category; 2) Add a date range filter; 3) Clear filters and open page 2	Category Music, a date range of the next 8 days	Filters combine correctly, each page has at most 10 events and page 2 shows the next ones; cancelled and past events are not listed, and a cancelled event's own page shows "Status: CANCELLED" (Updated after Sprint 2.)	Category filter: 12 events; category + date range (next 8 days): 7; no filters: page 1 has 10, page 2 has 4 of 14; the cancelled event is not in any result page, and its own page shows "Status: CANCELLED"	Pass
TC-EVT-10	F2 Event Management (I)	Attendee cannot create or publish events (EVT-F-010, EVT-F-005)	Logged in as Attendee	1) Attempt to create an event; 2) Attempt to publish an event	Attendee token	Both are refused with HTTP 403 and nothing is saved	Create: 403 FORBIDDEN: "You do not have permission for this action"; publish: 403 FORBIDDEN: "You do not have permission for this action"; nothing saved, the draft stays DRAFT	Pass

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-NFR-02	F2 Reliability and Compatibility (S)	Monthly availability (EVT-NF-002)	Health-check pinging every minute is set up	1) Collect health-check results for the test period; 2) Calculate uptime excluding announced maintenance	Health check every 1 minute	Uptime is 99% or higher	Uptime monitor (backend/scripts/uptime-monitor.js) pinged /health 10 times, 6 s apart: 10/10 up, 100%. After a Docker engine restart on 10 Nov the backend came back by itself (restart policy). A full month needs a deployed server; see docs/test-reports/nf-002-availability.md	Pass (short test period)
TC-NFR-06	F2 Reliability and Compatibility (S)	Browser compatibility (EVT-NF-006)	Latest two versions of Chrome, Firefox and Edge installed	1) Run the main flows (login, create event, search, register) in each browser	All three browsers	All acceptance tests pass on each browser with no layout or functional defect	Chrome (Chromium 153): login, create event, search, register and all other main flows pass with no layout or functional defect. Firefox and Edge were not tested	Partially executed (Chrome only)

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-SEC-03	F2 Security (I)	Input is validated and queries are parameterized (EVT-SR-003)	Logged in as Organizer	1) Enter ' OR 1=1 -- in login, search and event title fields; 2) Enter <script>alert(1)</script> in title and description; 3) View the saved event page	' OR 1=1 --; <script>alert(1)</script>	Injection strings are rejected or stored harmlessly; no script runs and no extra data is returned	Login with the injection string: 401 INVALID_CREDENTIALS: "Invalid email or password"; search: 0 results; an event titled "<script>alert(1)</script>' OR 1=1 --" is stored as plain text, and its page shows the text without running any script (checked in Chrome: the title shows as text and no alert appears); no tables changed	Pass

A.3 F3 Event Registration and Ticketing

Owner: Vishruta (PES1UG24CS538)

Covers: EVT-F-020 to EVT-F-027, EVT-NF-003, EVT-NF-004, EVT-SR-004, EVT-SR-005

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-REG-01	F3 Registration and Ticketing (S)	Register for an event with a free seat (EVT-F-020)	Published future event with free seats; logged in as Attendee	1) Open the event; 2) Click Register	Event capacity 50, 10 registered	Registration is Confirmed, free seats drop by 1 and a unique ticket ID is shown	Status CONFIRMED, free seats 40 → 39, ticket ID TKT-B71608564D shown	Pass
TC-REG-02	F3 Registration and Ticketing (I)	Registration refused for Draft, Cancelled or started events	A Draft event, a Cancelled event and an event whose start time	1) Try to register for each of the three events	Three events	Each attempt is refused with a message and no registration is created	Draft: 400 EVENT_NOT_OPEN: "This event is not open for registration"; Cancelled: 400 EVENT_NOT_OPEN: "This event is not open for	Pass

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
		(EVT-F-020)	has passed exist				registration"; Started: 400 EVENT_NOT_OPEN: "This event is not open for registration"; no registration created	
TC-REG-03	F3 Registration and Ticketing (I)	Prevent duplicate active registration (EVT-F-021)	Attendee already has a Confirmed or Waitlisted registration for the event	1) Try to register again for the same event	Same attendee and event	Message "Already registered" is shown and no second registration is created	409 ALREADY_REGISTERED: "You are already registered for this event"; still one registration	Pass
TC-REG-04	F3 Registration and Ticketing (S)	Join the waitlist when the event is full (EVT-F-022)	Event has capacity 2 and 2 Confirmed registrations	1) Register a third and then a fourth attendee	Attendees C and D	Both become Waitlisted, with positions 1 and 2 shown	C: WAITLISTED position 1; D: WAITLISTED position 2	Pass
TC-REG-05	F3 Registration and Ticketing (S)	Cancel own registration (EVT-F-023)	Attendee has a Confirmed registration for a future event	1) Open My registrations; 2) Cancel the registration; 3) Check the event's free seats	One Confirmed registration	Status becomes Cancelled and free seats increase by 1	Status CANCELLED; free seats 4 → 5	Pass
TC-REG-06	F3 Registration and Ticketing (I)	Waitlist promotion in first-in-first-out order (EVT-F-024)	Full event with waitlist C (position 1) and D (position 2)	1) A Confirmed attendee cancels; 2) Check statuses of C and D and the notifications	Waitlist C then D	C becomes Confirmed, D moves to position 1 and C is notified	A cancels → C CONFIRMED, D Waitlisted at position 1; C has a PROMOTED in-app notification and the email "A seat is free, you are confirmed: Waitlist test"	Pass
TC-REG-07	F3 Registration and Ticketing (S)	My registrations list (EVT-F-025)	Attendee has Confirmed, Waitlisted and Cancelled registrations; another attendee also has registrations	1) Open My registrations	Three registrations of one attendee	Only this attendee's registrations are listed, each with event name, date and the correct status	3 registrations listed, all belonging to this attendee (database count 3), each with event name, date and status (CANCELLED, CONFIRMED, WAITLISTED); other attendees' registrations are not shown	Pass
TC-REG-08	F3 Registration and Ticketing (S)	Ticket page with QR code (EVT-F-026)	Attendee has a Confirmed registration	1) Open the ticket page; 2) Scan the QR code with a scanner or phone; 3) Open the ticket page of a Waitlisted registration	One Confirmed ticket	Ticket page shows ID, event details and QR code; the scan returns the same ticket ID; a Waitlisted registration has no ticket	Ticket page shows TKT-B71608564D, "Reg test", date, venue and a QR code; decoding the QR image returns TKT-B71608564D (same ID); Waitlisted registration: 409 INVALID_STATE: "A ticket is only available for a confirmed registration", and My registrations shows no "View ticket" button for it	Pass
TC-REG-09	F3 Registration and Ticketing (S)	Organizer views the registration list (EVT-F-027)	Organizer owns an event with registrations; a second organizer exists	1) Open the registration list of the own event; 2) As the second organizer try to open the same list	Event with 5 registrations	The list shows name, email, status and registration time; the other organizer is refused with HTTP 403	List shows 4 registrations with name, email, status and registration time; second organizer: 403 FORBIDDEN: "You can only view registrations for your own events"	Pass
TC-REG-10	F3 Registration	Cancel after the	Attendee has a	1) Try to cancel the	Started event	Cancellation is refused	400 EVENT_NOT_OPEN:	Pass

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
	and Ticketing (I)	event has started is refused (EVT-F-023)	Confirmed registration for an event whose start time has passed	registration		with a message and the status stays Confirmed	"Registrations cannot be cancelled after the event has started"; status stays CONFIRMED	

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-NFR-03	F3 Usability and Concurrency (S)	Usability and responsive layout (EVT-NF-003)	Published event and 5 test users	1) Ask each user to register for an event starting from its details page and count the clicks; 2) Check every page at 360 px and 1920 px width	5 users, widths 360 px and 1920 px	Registration needs at most 3 clicks and every page displays correctly at both widths	A visitor needs 3 clicks from the event page (Log in, Login, Register) and is returned to the event after logging in; a logged-in attendee needs 1. 13 pages (events, event page, login, sign-up, my registrations, notifications, profile, my events, create, edit, registrations, report, check-in) checked at 360 px and 1920 px in Chrome: no horizontal scrolling, no overlapping text	Pass
TC-NFR-04	F3 Usability and Concurrency (I)	No overbooking under 500 simultaneous attempts (EVT-NF-004)	Event with capacity 100 and 500 test accounts	1) Run a script that sends 500 simultaneous registration requests; 2) Count Confirmed registrations; 3) Repeat the run 3 times	Capacity 100, 500 accounts	Confirmed registrations never exceed 100 in any run and the extra attendees are Waitlisted	backend/scripts/concurrency-test.js with capacity 100 and 500 simultaneous requests, run 3 times: every run 500 x HTTP 201, exactly 100 Confirmed and 400 Waitlisted, in 3.1 s, 2.1 s and 2.2 s. Never overbooked	Pass

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-SEC-04	F3 Security (S)	HTTPS and TLS version (EVT-SR-004)	Application deployed on the test server	1) Connect with openssl s_client using TLS 1.0, 1.1, 1.2 and 1.3; 2) Open the site over plain HTTP (Updated after Sprint 2.)	https://localhost:8443 and http://localhost:8080	No protocol below TLS 1.2 is offered and plain HTTP is redirected to HTTPS or refused	TLS 1.0 and TLS 1.1 refused with a "protocol version" alert; TLS 1.2 and TLS 1.3 negotiated; http://localhost:8080/events/1 answers 301 to https://localhost:8443/events/1	Pass
TC-SEC-05	F3 Security (I)	Ownership checks on registrations and tickets (EVT-SR-005)	Attendee A and attendee B each hold a registration; organizer of another event exists	1) As A request B's registration by ID; 2) As A request B's ticket by ID; 3) As an organizer request registrations of another organizer's event	IDs belonging to other users	Every request returns HTTP 403 and no data is returned	A requesting B's ticket: 403 FORBIDDEN: "You can only view your own tickets"; A cancelling B's registration: 403 FORBIDDEN: "You can only cancel your own registrations"; organizer requesting another organizer's registrations: 403 FORBIDDEN: "You can only view registrations for your own events"; no data returned (there is no endpoint	Pass

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
							that returns another user's registration by ID)	

A.4 F4 Check-in, Notifications and Reports

Owner: Nithin Kumar M (PES1UG25CS829)

Covers: EVT-F-030 to EVT-F-037, EVT-NF-005, EVT-SR-006, EVT-SR-007

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-CHK-01	F4 Check-in, Notifications and Reports (S)	Check in a valid ticket (EVT-F-030)	Organizer owns the event; attendee has a Confirmed ticket	1) Open the check-in screen; 2) Scan or type the ticket ID	Valid ticket ID	Ticket becomes Checked-in with a timestamp and the attendee name is shown	200; screen shows "Checked in: Check One (TKT-04A3F833A0)"; database status CHECKED_IN with timestamp 2026-11-10T06:09:26.433Z	Pass
TC-CHK-02	F4 Check-in, Notifications and Reports (I)	Reject an invalid, cancelled or waitlisted ticket (EVT-F-031)	Own event with a Cancelled and a Waitlisted registration	1) Enter a ticket ID that does not exist; 2) Enter the Cancelled ticket; 3) Enter the Waitlisted ticket	Unknown ID, Cancelled ticket, Waitlisted ticket	Each case shows its own message and no check-in is recorded	unknown: 404 NOT_FOUND: "Invalid ticket: no ticket with this ID exists"; cancelled: 409 INVALID_STATE: "This registration was cancelled"; waitlisted: 409 INVALID_STATE: "This attendee is on the waitlist and has no confirmed seat"; no check-in recorded	Pass
TC-CHK-03	F4 Check-in, Notifications and Reports (I)	Reject a used ticket or a ticket for another event (EVT-F-031)	A ticket already checked in; a ticket of a different event	1) Check in the same ticket a second time; 2) Enter the other event's ticket	Used ticket, other-event ticket	Each case shows its own message and no new check-in is recorded	Second check-in: 409 TICKET_ALREADY_USED: "Ticket already used: checked in at 10 Nov 2026, 11:39 am"; other event's ticket: 404 NOT_FOUND: "This ticket belongs to a different event"; no new check-in	Pass
TC-CHK-04	F4 Check-in, Notifications and Reports (I)	Emails for the four registration situations (EVT-F-032)	Mock SMTP service running; event and attendee exist	1) Register (Confirmed); 2) Promote from the waitlist; 3) Cancel a registration; 4) Cancel the event	One attendee per situation	One email is sent for each situation, containing the event name and the new status	MailHog/mock SMTP received exactly one email each: "Registration confirmed: Checkin test" (1), "A seat is free, you are confirmed: Checkin test" (1), "Registration cancelled: Checkin test" (1), "Event cancelled: Cancel test" (1); each contains the event name, date, venue and status	Pass
TC-CHK-05	F4 Check-in, Notifications and Reports (I)	Reminder email 24 hours before the event (EVT-F-033)	Event starting in 24 hours with 3 Confirmed attendees and 1 Waitlisted attendee	1) Run the scheduled reminder job; 2) Count the emails sent to each attendee	3 Confirmed, 1 Waitlisted	Each Confirmed attendee gets exactly one reminder within plus or minus 5 minutes of the 24-hour mark and the Waitlisted attendee	Reminder job (runs every minute) sent exactly one reminder each to the 3 Confirmed attendees (1, 1, 1) and none to the Waitlisted one (0), also after a second run; sent 1438.3 minutes before the start (24-hour mark = 1440)	Pass

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
						gets none		
TC-CHK-06	F4 Check-in, Notifications and Reports (S)	In-app notification list (EVT-F-034)	User has received a notification	1) Open the notification list; 2) Mark one notification as read	Two notifications	New items appear as unread and change to read after marking; only the user's own notifications are shown	List shows 2 notifications, 2 unread (marked "New"); after "Mark as read": 1 unread and the item shows as read; another user marking it: 403 FORBIDDEN: "You can only change your own notifications"; each user sees only their own list	Pass
TC-CHK-07	F4 Check-in, Notifications and Reports (I)	Attendance report (EVT-F-035)	Event with 10 Confirmed registrations, 8 of them checked in, 2 Waitlisted, 1 Cancelled	1) Open the report as the event's Organizer	10 Confirmed, 8 Checked-in	Counts are correct and the attendance rate shows 80%	Confirmed 10, Checked in 8, Waitlisted 2, Cancelled 1, attendance rate 80%	Pass
TC-CHK-08	F4 Check-in, Notifications and Reports (S)	CSV export of the attendee list (EVT-F-036)	Own event with registrations	1) Click Export CSV; 2) Open the file in a spreadsheet	Event with 5 registrations	One row per registration with name, email, status and check-in time; a second organizer's export request is refused	File attendees-event-28.csv with header "Name,Email,Status,Checked in at" and 13 rows (one per registration, check-in time filled for the 8 checked in); opens in a spreadsheet; second organizer: 403 FORBIDDEN: "You can only see reports for your own events"	Pass
TC-CHK-09	F4 Check-in, Notifications and Reports (S)	Admin dashboard totals (EVT-F-037)	Logged in as Admin; test data in the database	1) Open the dashboard; 2) Compare each total with a database count query	Users by role, events by status, registrations	All dashboard totals equal the database counts	Dashboard = database: attendees 50=50, organizers 2=2, admins 1=1; events draft 4=4, published 23=23, cancelled 1=1; registrations 63=63	Pass
TC-CHK-10	F4 Check-in, Notifications and Reports (U)	Report with zero confirmed registrations (EVT-F-035)	Own event with no Confirmed registrations	1) Open the report	Event with 0 Confirmed	Counts show 0 and the attendance rate shows 0% without an error or a division by zero	200; all counts 0 and attendance rate 0% with no error	Pass

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-NFR-05	F4 Maintainability (S)	Unit test coverage and quality gate (EVT-NF-005)	CI pipeline is running on the backend	1) Run the backend unit tests with coverage (npm test); 2) Run ESLint (npm run lint); 3) Raise the coverage threshold above the current coverage and check that the test run fails (Updated after Sprint 2.)	Backend unit test suite	Line coverage is at least 70% and the test run (and so CI) fails below it; ESLint reports no errors (Updated after Sprint 2.)	205 backend tests pass; coverage 86.79% lines, 85.26% statements, 78.64% branches; ESLint 0 errors in backend and frontend; with the threshold raised to 95% Jest exits with "coverage threshold for lines (95%) not met", so CI would fail. SonarQube analysis was not run: the setup is in the repository, but the quality gate is enforced by ESLint and the coverage	Pass with deviation

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
							threshold in CI	

Test Case ID	Module	Description	Pre-conditions	Steps	Test Data	Expected Result	Actual Result	Result
TC-SEC-06	F4 Security (I)	Audit log entries (EVT-SR-006)	Audit logging enabled	1) Perform a login success and a login failure, a role change, an event cancellation and a check-in; 2) Open the audit log; 3) Search the log for passwords and tokens	One of each action	One correct entry (timestamp, user ID, action) exists for each action and no password or token appears in the log	audit_logs has entries with timestamp, user ID and action for LOGIN_SUCCESS, LOGIN_FAILED, ROLE_CHANGED, EVENT_CANCELLED, CHECK_IN (83 entries in total); searching every entry for the test passwords and for token text ("eyJ") finds nothing	Pass
TC-SEC-07	F4 Security (I)	Rate limiting (EVT-SR-007)	Rate limiting enabled	1) Send 101 API requests from one IP within a minute; 2) Send 11 login attempts within a minute	Request script from one IP	The 101st request and the 11th login attempt return HTTP 429	At the normal limits: API requests 1-100 got 200 and the 101st got 429 TOO_MANY_REQUESTS; login attempts 1-10 got 401 (wrong password) and the 11th got 429	Pass